TEST REPORT

Automotive

Test of:	Motor controller module.
Model Number:	Lincoln HVLP-10.
SGS Reference :	AUT206306/GH/15.
Applicant :	Sevcon Limited.
Test Specification:	ISO 16750-4: 2010* Clause 5.3.2, 5.5.1 & 5.7
Date of Receipt:	30th June 2015.
Date of Test(s):	1st July to 24th August 2015.
Date of Issue:	25th August 2015.
Issue No:	One.
Conclusion :	The sample was tested to comply with the standard as per clients instructions.

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Tests marked:"*" are not covered by our UKAS accreditation.

Signature  Signature 

Test Engineer G. Hann

Authorised Signatory F. Huggins ~~A. Reynard G. Hann~~

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1. Client Information

Company Name:	Sevcon Limited.
Address:	Kingsway South, Team Valley Trading Estate, Gateshead, Tyne & Wear, NE11 0QR.
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2. Equipment Under Test (EUT)

2.1 Identification of EUT

Model Number:	Lincoln HVLP-10. [also PCB 8924175000110-1452]
Unique Identifier:	1504100022.
Description of EUT:	Motor controller.
Supply Voltage:	12 VDC.
Accessories Supplied:	Client supplied motors and loads.

3. Purpose of Tests

To perform the relevant tests and assess the product for compliance with the above specification.

4. Deviations or Exclusions from the Test Specifications

There were no deviations from the test specifications.

The scope of the inspection is limited to what is specified in the clients' instructions and does not include any other checks or tests.

5. Operation of the EUT During Testing

5.1 Configuration and Peripherals

Peripheral or support equipment was required for the tests.

5.2 Operating Mode and Environmental Conditions:

The operating modes and environmental conditions used for each individual test are described in the test results section of this report.

Methods and Procedures

Tests were carried out in accordance with the methods and procedures of the following test specifications.

Standard	Date	Description	Applicable
ISO 16750 – 4.	2010	Clause 5.3.2. Rapid temperature change.	Yes
ISO 16750 – 4.	2010	Clause 5.5.1. Salt mist test, severity 5.	Yes
ISO 16750 – 4.	2010	Clause 5.7. Damp heat, steady state test.	Yes

Operating Environment

Power Supply:	12 Volts DC	
Environmental conditions:-		
Temperature:	^{o}C	18.5 – 23
Relative Humidity:	%	41 - 55
Barometric Pressure:	m-Bar	998 - 1021

Operating Modes:

Mode “motor running” “motor not running” & “motor not connected”.

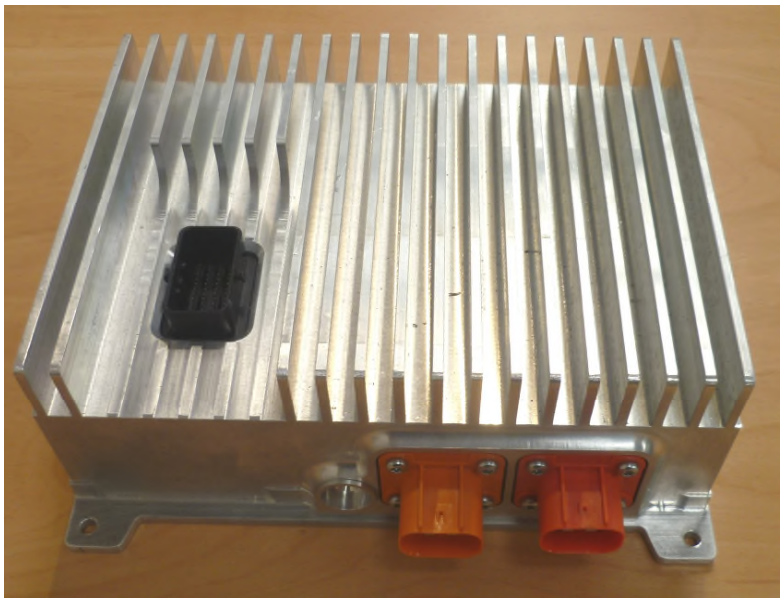
Configuration & Peripherals

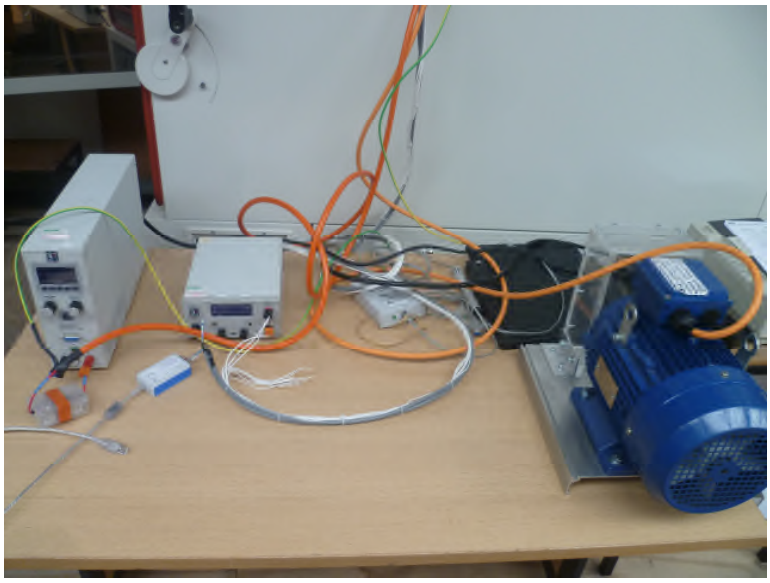
Peripherals were attached to the EUT during testing.

Summary of Test Results

Standard	Description	Result
ISO 16750 – 4. EN 60068-2-14.	Clause 5.3.2. Rapid temperature change.	Complied.
ISO 16750 – 4. IEC 60068-2-52.	Clause 5.5.1. Salt mist test, severity 5.	Complied.
ISO 16750 – 4. IEC 60068-2-78.	Clause 5.7. Damp heat, steady state test.	Complied.

Photograph of the EUT.





Clause 5.3.2: [Rapid Temperature change].

See EN 60068-2-14: Test Na. "With the specimens un-powered".

Sample: [PCB 8924175000110-1452]

PARAMETERS	
Upper temperature	+85° C ± 2° C
Lower Temperature	-40° C
Upper temperature time	60 minutes
Lower temperature time	60 minutes
Change over time per cycle.	Less than 2minutes.
Total number of cycles	100.

Procedure:
The system shall be in the un-powered state for the 100 cycles.
NB. Visually Inspect the PCB: components/soldering ETC after 100cycles for any deviations form new/start.

EUT shall not be powered and its "case/housing removed for the test.
The client done functional checks on the PCB; before starting the test and will do them again on completion of the testing.

Equipment:
K. Coupler "Reel 15K – 11" & Thermocouple RS 206-3738 S/No. 63101925.
Freezer: Frigor Electronic. S/No: 706350003.
K. Coupler "L5 – 7" & Thermocouple RS 206-3738 S/No. 63101925.
Oven: Gallenkamp – Oven BS, Size one. S/No. 20208.

NB. The client supplied the "PCB" sample and on completion of the test the client checked the operation of the sample.

Cold temperature range: -39.1˚C to -40.2˚C.
Hot temperature range: +84.6˚C to +85.2˚C.

Clause 5.3.2: [Rapid Temperature change].

Start.





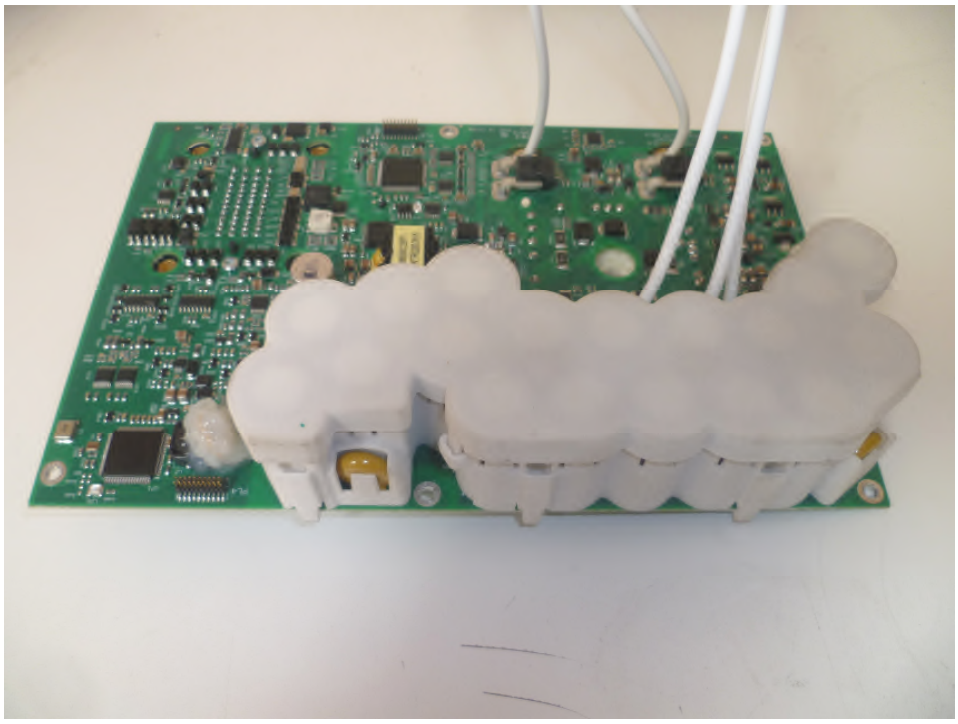




Clause 5.3.2: [Rapid Temperature change].

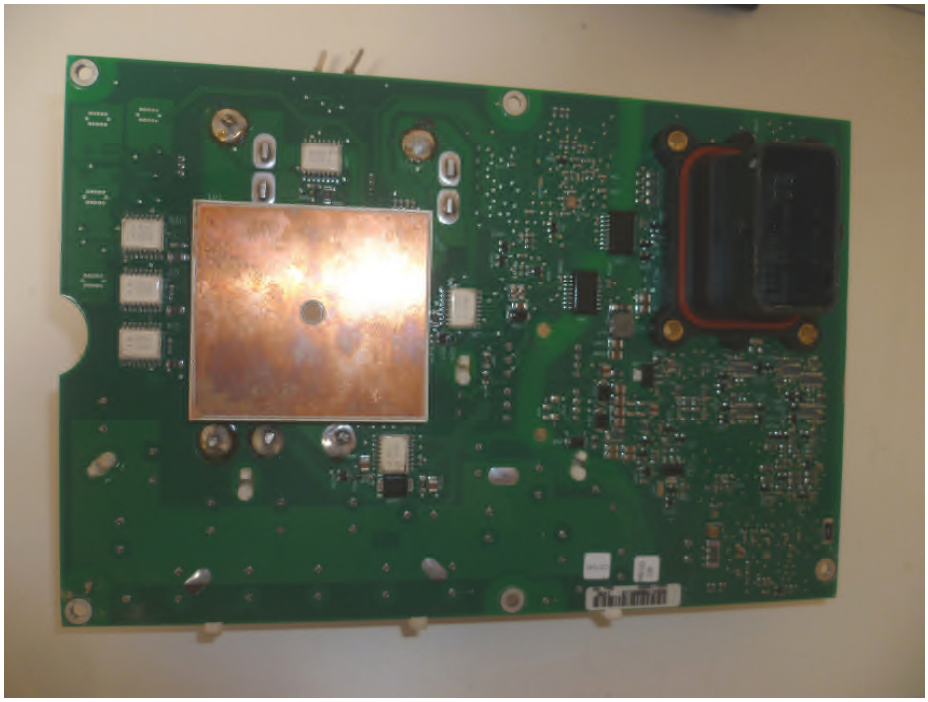
Photo after 100 test cycle.





Clause 5.3.2: [Rapid Temperature change].

Photo after 100 test cycle.





Clause 5.7: [Damp Heat steady state].

See IEC 60068-2-78: Test with the specimens powered.

PARAMETERS	
Upper temperature	+40° C ± 2° C
Humidity at upper temperatures	85% ± 3% RH
Duration "Days"	21
Operating mode: 2.1 (Sleep mode).	First 20 days and 23 hours.
Operating mode: 3.2 (Operating mode).	Last hour.

Procedure

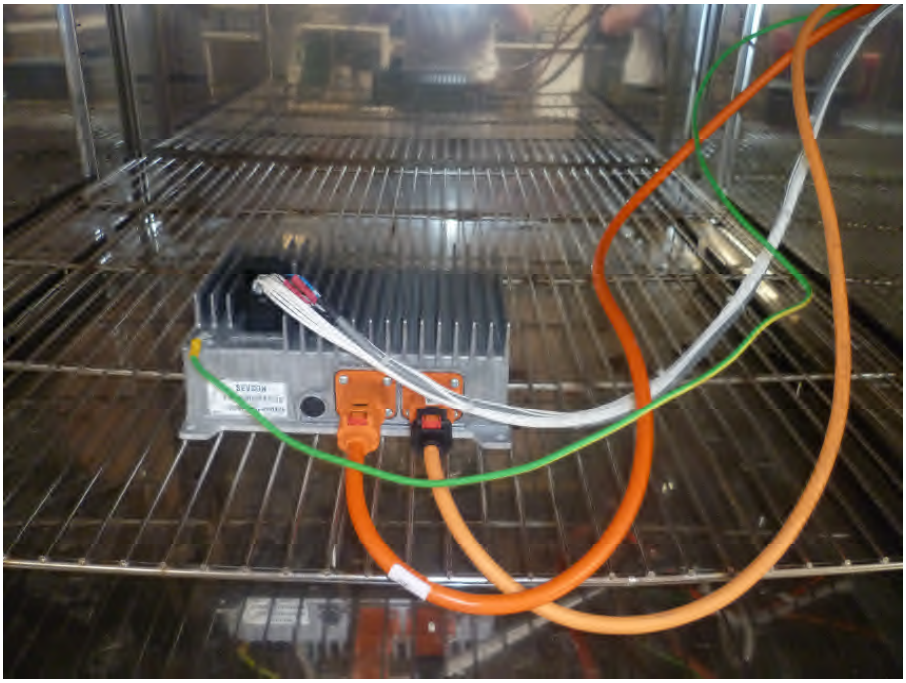
1) Place the system in the condition at the start of the test, and after each 20 days and 23 hours change operating mode from 2.1 to 3.2 for the last hour.

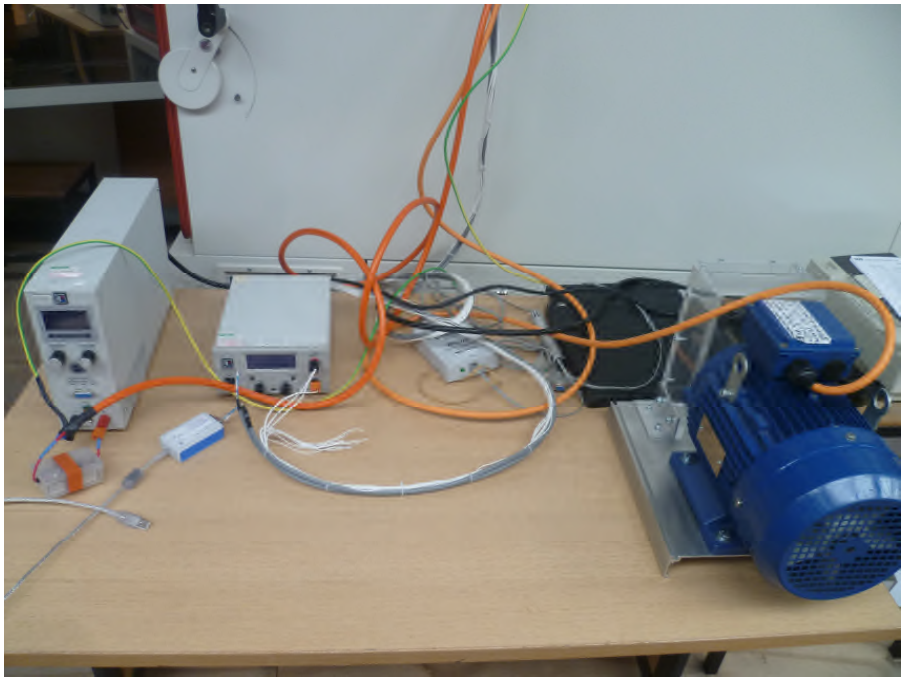
2) After a recovery period of 0.5 hour at the end of the test, the client will carry out his tests.

Temperature range: +39.9˚C to +40.1˚C.
Humidity range: 84.7% to 85.1%.

Clause 5.7: [Damp Heat steady state].

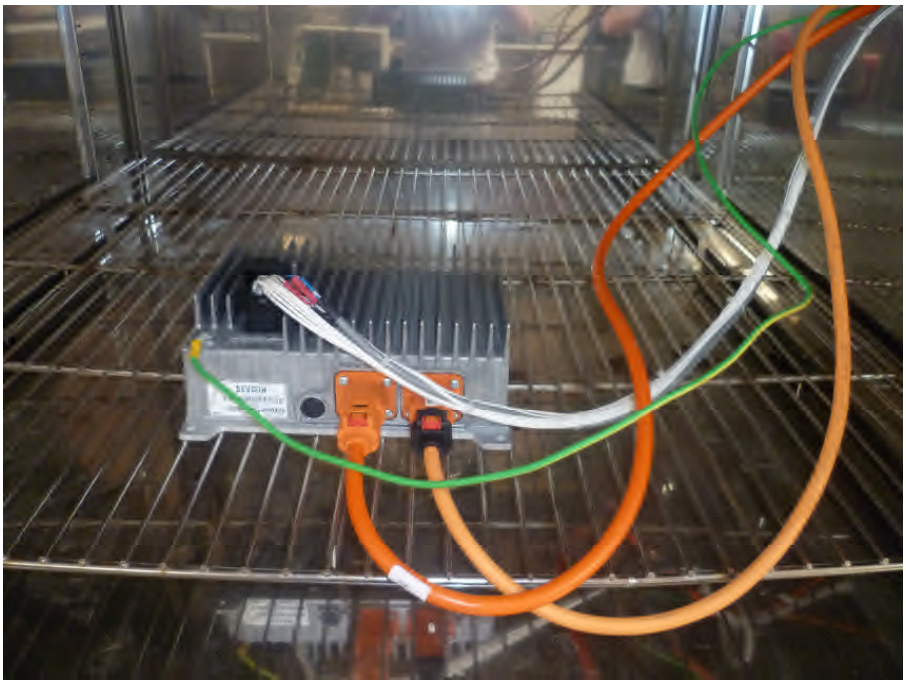
Photo of EUT in the chamber at start of tests:





Clause 5.7: [Damp Heat steady state].

Photo of EUT in the chamber at end of tests:



On completion on the test cycles the client powered up the sample and proved it worked correctly,

Then clauses 4.11 and 4.12 where conducted using:

CLARE et/s 011:1 Model A203D/213/500VA/SP. S/No. 76217A1.

(1) 500v RMS 50-60Hz for 1 min was applied with no break down.

(2) 500v DC for 1 min was applied with a resistance greater then 10M ohms was recorded on the connectors.

(3) Lab temperature 19.5; Humidity 55%; m-Bar 1010.

Clause 5.5.1: [Salt mist].

IEC 60068-2-52 Test Kb (Severity 5).
During conditioning the unit/system shall not be powered.

Apply the following conditioning cycle “4 times”.

Salt laden atmosphere.	
Concentration NaCl	5 +/- 1 %
pH @ 35 +/- 2°C	6.5 to 7.2
Temperature	+35 +/- 2°C
Humidity	› 85% RH
Duration	2 Hrs.
Temperature	+40 +/- 2°C
Humidity	93% RH
Duration	22 Hrs.
Duration - Lab	3 days +23°C / 45 to 55%

Equipment:
Climatic Chamber Vötsch VCS 4100.
Salt Mist Chamber Ascott 450L.
pH meter Piccolo.

Salt Mist Chamber:
pH range: 6.73 to 7.1

Climatic chamber:
Temperature range: +39.9˚C to +40.1˚C.
Humidity range: 92.8% to 93.1%.

Lab:
Temperature range: +20.5˚C to +23.0˚C.
Humidity range: 49% to 55%.

Clause 5.5.1: [Salt mist].

Photo Start:



Photo End:



Clause 5.5.1: [Salt mist].

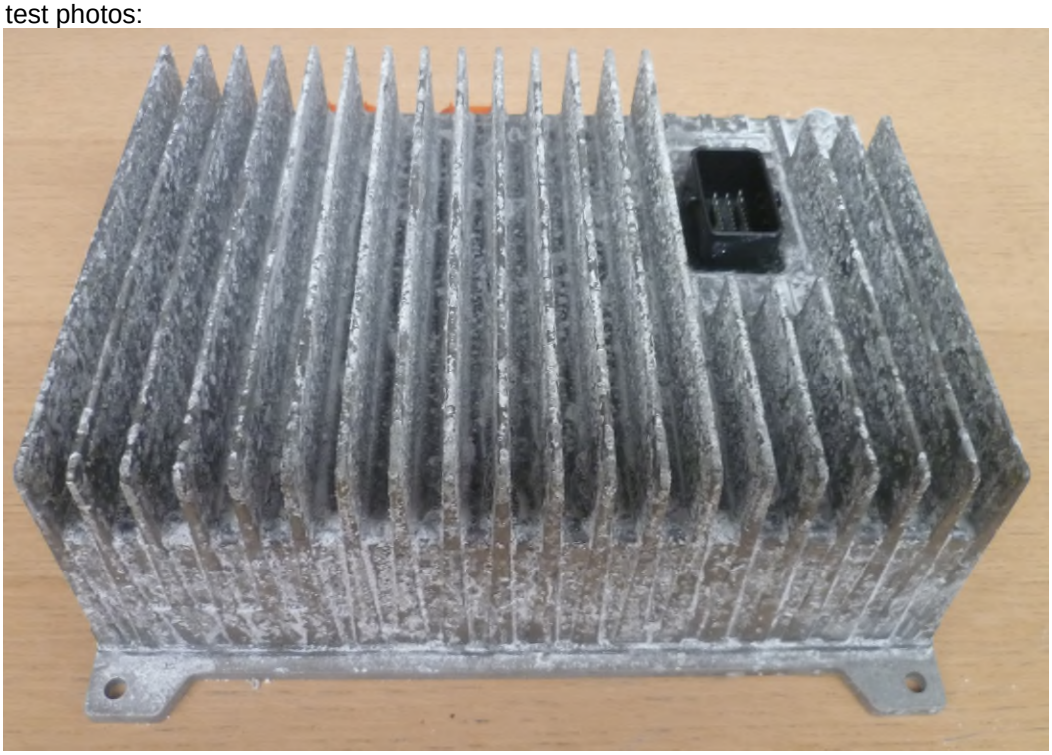
End of test photos:

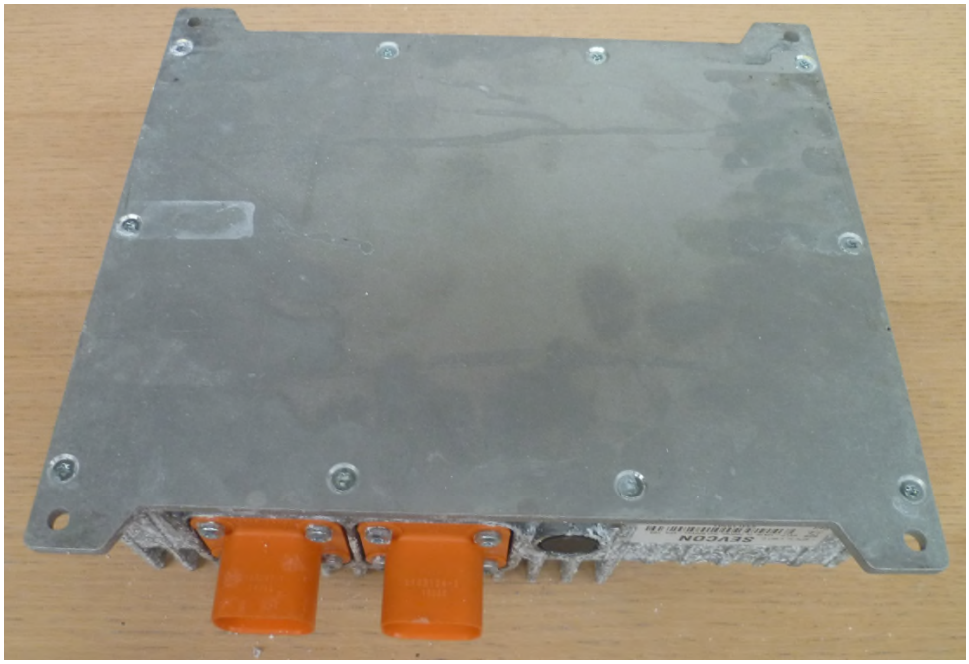




Clause 5.5.1: [Salt mist].

End of test photos:





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